

Final Project: Visual Understanding with TextVQA

AI in Medicine - Spring 2026

Due: May 7, 2026, 11:59 pm

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1 Project Overview

This final project focuses on visual understanding using the TextVQA dataset. You will train and evaluate vision language models (VLMs) to understand text embedded in images and answer questions correctly.

Team Requirements: This project will be conducted in groups with a maximum of four students per team.

2 Dataset

2.1 TextVQA Dataset Description

The TextVQA dataset is designed to test models' ability to read and reason about text in natural images. It contains:

- **34.6k/5k/5.73k** paired QA-image samples for **train/validation/test**
- Questions require understanding text embedded in real-world scenes
- Examples include reading airplane tail text, phone displays, measuring cups, currency denominations, etc.

2.2 How to Access the Dataset

The TextVQA dataset can be accessed through the following methods:

1. **Hugging Face Datasets:** <https://huggingface.co/datasets/lmms-lab/textvqa>
 - Download the dataset directly from the official repository
 - Includes training, validation, and test splits

Question: “What is the breed of phone?”

Answers: “nokia”

Image ID: 0046c5197f2f7e4e

flickr_300k_url:

https://farm6.staticflic...f65b421097_o.jpg



Figure 1: Example from TextVQA dataset. This example demonstrates that the model must read the text “NOKIA” visible on the phone to correctly answer the question.

2. Dataset Structure:

- Images in JPEG format
- Annotations in JSON format containing questions, answers, and OCR tokens
- Train/Validation/Test splits are pre-defined

3 Recommended Models

You can use one of the following fully open-source pretrained VLMs, though other models are also acceptable and welcome.

1. BLIP-2 with OPT-2.7B

- Access via Hugging Face: `Salesforce/blip2-opt-2.7b`
- GitHub Repo: <https://github.com/salesforce/LAVIS/tree/main/projects/blip2>
- Parameters: 2.7B
- Efficient image-text understanding with Q-Former architecture

2. MiniGPT-4

- Access via Hugging Face: `Vision-CAIR/MiniGPT-4`
- GitHub Repo: <https://github.com/Vision-CAIR/MiniGPT-4>
- Parameters: 7B
- Inspired by GPT-4’s vision capabilities

3. LLaVA-Phi-3-mini

- Access via Hugging Face: `xtuner/llava-phi-3-mini`
- Parameters: ~ 3.8 B
- Lightweight LLaVA variant with strong visual reasoning

4. Qwen2.5-VL-3B

- Access via Hugging Face: `Qwen/Qwen2.5-VL-3B-Instruct`
- Parameters: $\sim 3\text{B}$
- Multilingual vision-language model

5. InternVL2.5

- Access via official website: <https://internvl.github.io/blog/2024-12-05-InternVL-2.5/>
- Parameters: $1\text{B} \sim 78\text{B}$
- Multilingual vision-language model

6. Other VLMs

- Search online and pick one of them
- Parameters: $\leq 32\text{B}$

Note: All recommended models are fully open-source and designed to run efficiently on limited computational resources.

4 What to do

4.1 Task Description

Your task is to evaluate and fine-tune a VLM on TextVQA dataset, including three sub-tasks below:

1. **Zero-shot Evaluation:** Evaluate pretrained models directly on TextVQA valid/test sets
2. **Fine-tuning or Prompt Engineering** (use one of them)
 - **Fine-tuning:** Fine-tune the selected pretrained model with LoRA on the TextVQA training set
 - **Prompt Engineering:** Design and optimize prompts to improve reasoning and answer accuracy without modifying model parameters.
3. **Result Analysis:** Compare zero-shot and fine-tuned (or prompt-engineered) results; report accuracy and qualitative examples; analyze common error types (e.g., OCR misreads, reasoning failures) to identify model limitations and improvement directions.

4.2 Required Components

Your implementation and report should include:

- Data loading and preprocessing pipeline
- Model initialization and configuration
- Training loop (if fine-tuning) or prompt engineering details
- Inference and evaluation code
- Results visualization and analysis

4.3 Evaluation Metrics

You must evaluate your model(s) using the following metrics:

- **Accuracy (Primary)** Percentage of correctly answered questions
 - TextVQA uses exact string matching with answer list
 - Report accuracy on validation and/or test sets
- **Semantics (Secondary)** Percentage of correctly answered questions
 - BLEU
 - METEOR
 - ROUGE
 - LLM-as-a-Judge similarity
- **Additional Metrics (Optional but Recommended)**
 - F1 Score: Token-level overlap between prediction and ground truth
 - Precision/Recall: For substring matching
 - Per-Category Performance: Break down results by question types

5 Grading Criteria (50 points)

In general, report completeness, report quality, and model performance will serve as the three overarching grading criteria. Specifically, your final project will be evaluated based on the following aspects.

- **Introduction and Motivation and Background (5 points)**
 - Clear problem statement
 - Survey of relevant multimodal models

- **Methodology (15 points)**
 - Clear description of approach
 - Implementation details
 - Justification of design choices
 - Reproducibility (code, hyperparameters, etc.)
- **Experimental Design (15 points)**
 - Comprehensive evaluation setup
 - Proper train/val/test splits
 - Ablation studies well-designed and executed
- **Results and Analysis (10 points)**
 - Clear presentation of results (tables, figures)
 - Thorough analysis of findings
 - Discussion of failure cases
 - Comparison with baselines
- **Writing Quality (5 points)**
 - Clear and professional writing
 - Proper formatting and organization
 - Correct citations and references